Automatic Music Transcription (AMT)- Literature Review, CSES ORCA

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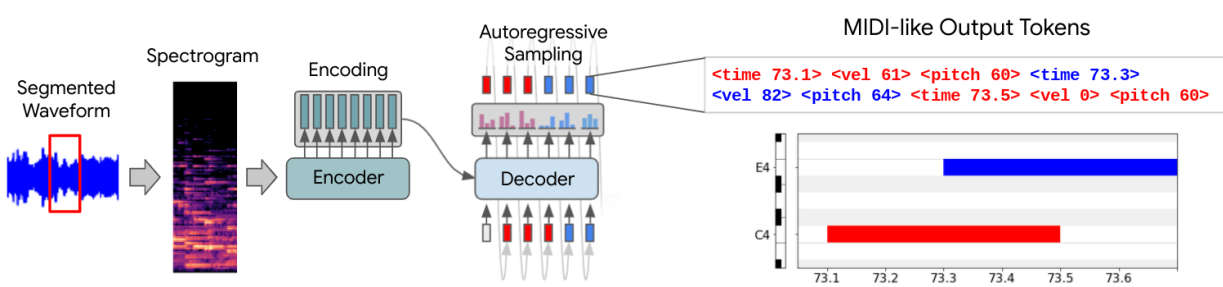
1 Introduction

AMT or Automatic Music Transcription is the process in which audio is automatically converted into symbolic representations like MIDI or sheet music. It can be leveraged in music education to make corrections, ease the friction between ideation and music creation, music searching, the study of genres, etc. The task of AMT is often split into distinct parts: Multi-pitch detection, Note onset/offset detection, Loudness estimation and quantization, Instrument recognition, Extraction of rhythmic information, Time quantization, Extraction of velocity and dynamic [1].

2 General Form

AMT's general pipeline as shown in Figure 1, involves converting a segmented waveform into a log-mel-spectrogram (frequency over time) often with a Short-Time Fourier Transform. It is the standard feature representation used by models and highlights musical information such as pitch and loudness. A sequence to sequence, transformer-based architecture processes this data. The encoder extracts a high-level content aware acoustic representation of music. It understands what is being played– pitches, timing and dynamics. The decoder is an autoregressive model that takes the encoded representation from the encoder and generates the output sequence (Midi-like Tokens), that can then be converted into MIDI or sheet music.



AMT System [2]

3 Major Techniques

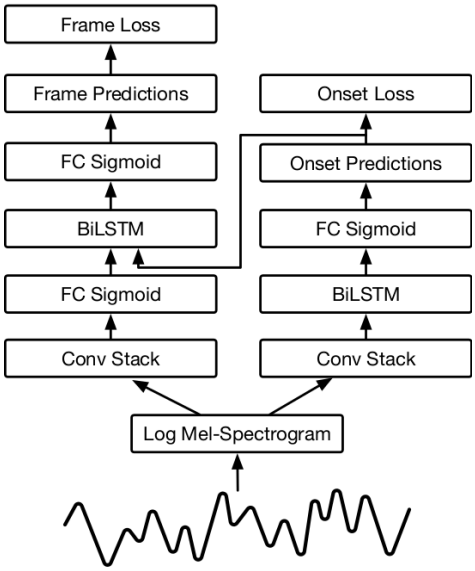
Non-negative Matrix Factorization (NMF) and Neural Networks (NNs) are two common approaches to AMT. Neural Network techniques typically involve processing spectrograms with various neural network architectures, such as long short-term memory layers or Convolutional Neural Networks (CNNs). [1] NMF involves decomposing the music spectrogram into a set of basis spectral templates (which can represent instruments or notes) and their corresponding temporal activation coefficients.

4 Key Models: Onset and Frames, MT3, MERT (Strengths, Weaknesses, and Opportunities)

Google Magenta's Onset and Frames (OaF) is one of the best baselines for piano transcription. Its design, see Figure 2, separates onset and frame head to more accurately identify the start and duration of a note, as well as velocity (how hard the note was played). The onset detector predicts when the note starts, the Frame detector predicts whether a note is active in each frame and the Offset detector predicts when the note ends. It was trained using tensorflow on the MAESTRO piano dataset with a batch size of 8, a learning rate of .0006, and a gradient clipping L2-norm of 3.

Strengths: It achieved state of the art performance in polyphonic piano datasets like MAESTRO and MAPS with an onset F1 score of 96.72% on MAESTRO, outperforming the original Onsets and Frames system's ~94.80% onset F1. [5]

Weaknesses: A limitation is its concentration in piano and later drums that significantly degrades its performance when transcribing other instruments or multiple instruments at once. Further, Its use of a Bidirectional LSTM (Bi-LSTM) requires context from the past and future of the audio, making real-time applications slow.



Recent work has expanded the OaF model that enables the transcription of multiple instruments via tokens identifying instruments to their output. Magenta's MT3 (Multi-Task, Multi-Track Music Transcription) is an MT3 is a SOTA (state-of-the-art) model for multi-instrument automatic music transcription that treats transcription as a language modeling problem. Instead of predicting piano-roll outputs, MT3 generates sequences of MIDI-like event tokens, such as note onsets, offsets, pitch, timing, and instrument information.

Figure 2 [4]

While MT3 performs well overall, it divides audio into non-overlapping segments and transcribes each segment independently. This design can cause the model to lose long-term musical context, leading to instrument leakage.

To address this issue, MR-MT3 incorporates information from previous audio segments. It introduces a memory retention mechanism that summarizes tokens from past segments and uses this information when transcribing the current segment. During training, MR-MT3 also applies prior token sampling. This allows the memory to come from earlier segments within a limited time window rather than only the immediately preceding one. Additionally, token shuffling is used as a data augmentation technique to reduce sensitivity to fixed token ordering. These modifications help preserve musical continuity across segments and significantly reduce instrument leakage, improving the overall quality of multi-instrument transcriptions.

Despite its improvements over MT3, MR-MT3 still has some limitations. The addition of memory retention and prior token sampling increases model complexity and computational cost, making training and inference more expensive.

Another technique that has been seen recently is the usage of self-supervised learning models, SSL. However, this approach does not capture music-specific characteristics such as polyphony, pitch structure, and harmonic relationships. [3] This is where MERT, Music Representation Transformer, comes in. It is a large-scale self-supervised framework tailored explicitly to music audio. MERT uses two complementary teacher models: an acoustic teacher based on residual vector quantization (RVQ-VAE) to capture general acoustic patterns so it answers the question of "What does this sound like acoustically?". And a musical teacher based on the Constant-Q Transform (CQT) to inject pitch- and harmony-aware supervision, answering the question "What notes and harmonies are present?".[1]

MERT is able to learn rich music-aware representations that significantly outperform speech-oriented SSL models across a wide range of music information retrieval tasks. It outputs continuous embeddings, so it tells us what is happening musically at this moment. Therefore, while MERT can not be directly used for audio transcription, it provides a strong foundation for downstreaming tasks, such as AMT, because it supplies meaningful audio features. With this, MERT can be paired with a transcription model, as that model will now have an easier job, learns faster, and generalizes better since it already has pitch relations, harmonic context, polyphony.

A limitation of MERT is it is trained on short audio segments. However, automatic music transcription primarily relies on local acoustic cues such as pitch and onsets, making sliding-window inference a viable strategy to address this limitation. Furthermore, pre-training MERT at scale is

[1] https://proceedings.iclr.cc/paper_files/paper/2024/file/33dffa2e3d2ab74a783d1a8c292f66d9-Paper-Conference.pdf

expensive and may be insatiable for larger model sizes. Finally, MERT learns music structure partly from a representation designed for Western tonal music, which helps in many cases but may restrict its ability to generalize beyond those musical conventions.

5 Proposed Solution

Based on the strengths and limitations of existing AMT systems, we propose an encoder–decoder transformer-based transcription model that integrates self-supervised music representations. Specifically, we aim to use a pretrained music-aware SSL model such as MERT as the encoder to extract rich pitch-, harmony-, and polyphony-aware audio features, and pair it with a lightweight autoregressive decoder that predicts MIDI-like event tokens similar to MT3. This approach reduces the need for large labeled datasets while improving generalization across instruments and genres. To support multi-instrument transcription and reduce instrument leakage, the model can incorporate limited contextual memory across audio segments, inspired by MR-MT3, while maintaining manageable computational costs.

6 Ethical Considerations and Threats

Building a tool that can configure the exact musical notes being played, especially at a large scale like orchestral music, has ethical and legal implications including copyright infringement. Before releasing such a tool, as researchers we must take care to address these concerns in addition to acknowledging them. This is something our team will continue to keep in mind as we progress through this project.

7 Conclusion

Overall, current research in automatic music transposition tends to excel in a specific aspect of music, such as transcribing individual instruments, or handling particular genres and musical styles. These models achieve strong performance in their target domain, but they often struggle to generalize across diverse musical contexts. As a result, there remains a significant gap in developing a well-rounded transcription model capable of accurately transcribing arbitrary music audio across instruments, genres, and musical traditions.

8 References

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